



Indian Institute of Technology Madras

**Software Engineering Project
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**Coaching Institute Operations Platform
Milestone – 01**

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Coaching Institute Operations Platform

This milestone focuses on identifying system users and deriving structured user stories based on stakeholder interactions conducted with faculty, students, administrative staff, parents, and management representatives.

1. IDENTIFIED USERS

The Coaching Institute Operations Platform involves multiple stakeholders categorized based on their level of interaction and operational responsibility within the institute.

Primary Users

Student

Students are the core users of the platform. They interact with the system to check attendance, view academic performance, access study materials, and stay informed about schedules and assessments.

Faculty (Teacher)

Faculty members use the system to record attendance, update marks, manage assignments, and communicate academic updates to students and parents.

Secondary Users

Data Entry Operator / Administrative Staff

Administrative staff are responsible for maintaining student records, updating admission details, managing fee entries, and ensuring data accuracy within the system.

Director of the Coaching Centre

The director oversees overall institute operations. This role involves monitoring performance trends, reviewing attendance patterns, and evaluating fee collection and institutional effectiveness.

Tertiary Users

Parent

Parents primarily use the platform to stay informed about their child's attendance, academic performance, and fee-related updates. Their interaction focuses on monitoring and communication rather than daily operations.

2. CONSOLIDATED PAIN POINT ANALYSIS

Based on stakeholder interactions, the following recurring challenges were identified across different user roles:

SL NO	CATEGORY	PAIN POINT	AFFECTED USERS	PRIORITY
1	Communication	Delay or inconsistency in schedule updates	Student, Parent	High
2	Attendance	Manual attendance tracking and reporting difficulties	Faculty, Admin	High
3	Academic Monitoring	No structured system to track student performance over time	Student, Faculty	High
4	Examination	Students skipping tests without structured tracking	Faculty, Director	High
5	Administration	Student data stored across multiple sheets leading to inefficiency	Admin	High
6	Academic Promotion	Manual class updates required during yearly promotions	Admin	Medium
7	Fee Communication	Lack of clear and structured fee communication	Parent	High
8	Parent Engagement	Limited direct communication between teachers and parents	Parent	High
9	Content Delivery	Lack of concise revision summaries for lengthy academic sessions	Student	Medium
10	Personalization	Lack of summarized 10-minute revision gist content	Student	Medium
11	Automation	AI-based auto-correction of project work	Faculty, Student	Medium

Note: Pain points categorized as High priority were mapped directly to core functional requirements, while Medium priority concerns were considered as enhancement features for future iterations.

3. USER STORIES

The following user stories have been derived from the prioritized pain points identified during stakeholder interactions. Stories are grouped into functional epics to reflect logical system domains.

Epic 1: Communication & Notifications

(Derived from Communication and Parent Engagement pain points.)

1. As a student, I want to receive notifications whenever a class schedule is updated so that I do not miss sessions.
2. As a parent, I want to receive attendance updates so that I can ensure my child is attending regularly.
3. As a parent, I want transparent fee communication so that there is no confusion regarding payments.
4. As a parent, I want to receive structured academic updates from teachers so that I can understand my child's strengths and areas requiring improvement.

Epic 2: Attendance Management

(Derived from Attendance tracking and manual reporting challenges.)

5. As a student, I want to view my attendance record so that I can stay aware of my regularity.
6. As a faculty member, I want to record attendance digitally so that consolidated reports can be generated without manual effort.
7. As a director, I want consolidated attendance insights so that I can monitor overall attendance trends.

Epic 3: Academic Performance Monitoring

(Derived from Academic Monitoring and performance visibility gaps identified across stakeholders.)

8. As a student, I want to track my performance across multiple assessments so that I can identify recurring weak subjects.
9. As a faculty member, I want to update marks in a structured format so that student progress can be reviewed systematically.
10. As a faculty member, I want a centralized view of student performance so that I can provide targeted academic guidance.
11. As a parent, I want access to performance reports so that I can evaluate my child's academic improvement.

Epic 4: Examination Accountability

(Derived from Examination participation gaps and lack of structured test tracking.)

12. As a faculty member, I want to monitor examination participation records to identify absentee patterns.
13. As a director, I want visibility into examination participation trends so that accountability gaps can be addressed.

Epic 5: Administrative Efficiency

(Derived from Administrative inefficiencies related to data fragmentation and manual promotion updates.)

14. As an administrative staff member, I want all student data to be stored in a centralized system so that I do not need to switch between multiple files.
15. As an administrative staff member, I want to generate customized attendance reports so that parent requests can be handled efficiently.

16. As an administrative staff member, I want student promotions to update automatically so that manual re-entry of class details is avoided.

Epic 6: Learning Enhancement (Medium Priority)

(Derived from Content Delivery and Personalization-related improvement opportunities.)

17. As a student, I want access to digital study materials so that I can revise topics if I miss a session.
18. As a student, I want visibility of upcoming tests and assignments so that I can plan my preparation effectively.
19. As a faculty member, I want AI-based auto-correction support for project work so that evaluation time can be reduced.

The stakeholder interaction recordings referenced in this milestone can be accessed through the link below:

Google Drive Link: [📁 USERS_AUDIO](#)